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Onboarding devices in standalone non-public networks

Abstract

The present disclosure relates to supporting provisioning of a User Equipment (UE) with credentials to access a communication network, such as a Standalone Non-Public Network (SNPN). A credential server stores a binding associating UE device information, including an onboarding identifier, with network information for the SNPN. The credential server receives, from a network function in an onboarding network, a request for authentication of the UE that includes the onboarding identifier. The credential server responds with a message comprising the network information bound to the device information to enable provisioning of the credentials to the UE.

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Background/Summary

RELATED APPLICATIONS

(1) The present application claims priority to European Application No. EP 20382410.7, which was filed on May 15, 2020, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates generally to standalone non-public networks (SNPN) and, more particularly, to procedures for onboarding devices in a standalone communication network.

BACKGROUND

(3) The Fifth Generation (5G) standard developed by the Third Generation Partnership Project (3GPP) includes support for non-public networks (NPNs), which are defined as “a network that is intended for non-public use” in 3GPP Technical Specification (TS) 22.261, Section 6.25. In general, NPNs are intended for use by a single entity, such as a large enterprise. One application for NPNs is in industrial automation with requirements for very low latency. NPNs can be deployed as standalone networks referred to as standalone NPNs (SNPNs). NPNs. can also be hosted by Public Land Mobile Networks (PLMNs) or configured as a slice of a PLMN.

(4) Regardless of the deployment scenario, all UEs within the NPN need to be provisioned with credentials to access the NPN. The UEs are typically manufactured by third party manufacturers and are not initially provisioned with credentials for accessing the NPN. Rather, provisioning is usually done when the UEs are purchased by a subscription owner, which is typically the owner of the NPN. The 5G standards provides a general outline of procedures for provisioning UEs with credentials for accessing a NPN. Generally, the UE initially registers with an onboarding network, which initiates a provisioning procedure with the NPN. Once the UE is provisioned with credentials for the NPN, the UE deregisters from the onboarding network and establishes a connection with the NPN using the obtained credentials.

(5) While existing procedures define the parties involved in the onboarding process, there are many aspects of the onboarding process that remain undefined and/or unresolved. One problem is that the UE is not provisioned in advance with information about the NPN of the subscription owner. Therefore, procedures are needed for network selection and routing of traffic related to provisioning. Further, it is expected that UEs will not attempt to access NPNs not associated with the subscription owner, which would burden resources in other NPNs. Therefore, procedures are needed to prevent rogue or malicious UEs from attempting to access an NPN other than one authorized by the subscription owner. Finally, different NPNs may have different procedures for provisioning UEs to access the NPN. A method is needed for the onboarding network to determine what procedures to use to provision the UE.

SUMMARY

(6) The present disclosure relates to provisioning of a UE with credentials to access a communication network, such as a SNPN. According to an aspect of the present disclosure, a

Default Credential Server (DCS) maintains a binding of UE onboarding credentials and a UE identifier with network information for the authorized SNPN. Therefore, pre-provisioning of the network information at the time of manufacture is not required. Rather, the manufacturer can update the DCS with the network information for the UE when the UE is acquired by the SNPN. Storing the network information in the DCS avoids the need to flash the firmware when the UE is manufactured or reflash the firmware when the UE is acquired. To initiate the onboarding process, the UE 50 sends an onboarding request containing the UE identifier to a network node in an onboarding network and the network node queries the DCS to retrieve the network information. After obtaining the network information from the DCS, the onboarding network requests authorization from a provisioning server to initiate a provisioning procedure with the SNPN. The provisioning server verifies that the UE is authorized to access the SNPN and determines the type of provisioning procedure to use (e.g., control plane provisioning or user plane provisioning). If verification is successful, the provisioning server sends a response authorizing the onboarding network to initiate provisioning of the UE and indicating the type of provisioning procedure to use. The authorization procedure prevents rogue or malicious UEs from attempting to initiate a provisioning procedure with the SNPN without prior authorization.

(7) A first aspect of the disclosure comprises methods implemented by a credential server for providing network information for a communication network, e.g. SNPN, to an onboarding network to enable provisioning of credentials to the UE to access the communication network. The method comprises storing a binding associating device information for the UE and network information identifying the communication network. The device information includes an onboarding identifier for the UE. The method further comprises receiving, from a network function in an onboarding network, a message including the onboarding identifier for the UE. The method further comprises sending, to the network function in the onboarding network, the network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(8) A second aspect of the disclosure comprises methods implemented by a network node in an onboarding network for obtaining network information for a communication network, e.g., SNPN, from a DCS to enable provisioning of credentials to the UE to access the communication network. The method comprises sending a message including an onboarding identifier for the UE to a DCS. The method further comprises receiving, from the DCS, network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(9) A third aspect of the disclosure comprises methods implemented by a network node in an onboarding network to obtain authorization for provisioning of a UE with credentials to access a communication network, e.g., SNPN. The method comprises sending, to a network function, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The method further comprises receiving an onboarding authorization response from the network function in the communication network allowing or rejecting the onboarding request. The method further comprises initiating an onboarding procedure when the authorization response indicates that provisioning of the UE is allowed.

(10) A fourth aspect of the disclosure comprises methods implemented by a provisioning server in a communication network to authorize provisioning of a UE with credentials to access a communication network, e.g., SNPN. The method comprises receiving, from a network function in an onboarding network, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The method further comprises verifying the onboarding identifier provided in the onboarding authorization request. The method further comprises sending an onboarding authorization response to the network function in the onboarding

network allowing or rejecting the onboarding authorization request depending on verification of the onboarding identifier.

(11) A fifth aspect of the disclosure comprises a credential server (e.g., DCS) configured to provide network information for a communication network, e.g., SNPN, to an onboarding network to enable provisioning of credentials to the UE to access the communication network. The credential server is configured to store a binding associating device information for the UE and network information identifying the communication network. The device information includes an onboarding identifier for the UE. The credential server is further configured to receive, from a network function in an onboarding network, a message including the onboarding identifier for the UE. The credential server is further configured to send, to the network function in the onboarding network, the network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(12) A sixth aspect of the disclosure comprises a network node in an onboarding network configured to obtain network information for a communication network, e.g., SNPN, from a DCS to enable provisioning of credentials to the UE to access the communication network. The network node (e.g., AMF) is configured to send a message including an onboarding identifier for the UE to a DCS. The network node is further configured to receive, from the DCS, network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(13) A seventh aspect of the disclosure comprises methods implemented by a network node (e.g., AMF) in an onboarding network configured to obtain authorization for provisioning of a UE with credentials to access a communication network, e.g., SNPN. The network node is configured to send, to a network function, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The network node is further configured to receive an onboarding authorization response from the network function in the communication network allowing or rejecting the onboarding request. The network node is further configured to initiate an onboarding procedure when the authorization response indicates that provisioning of the UE is allowed.

(14) An eighth aspect of the disclosure comprises methods implemented by a provisioning server in a communication network, e.g., SNPN, configured to authorize provisioning of a UE with credentials to access a communication network, e.g., communication network. The provisioning server is configured to receive, from a network function in an onboarding network, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The provisioning server is further configured to verify the onboarding identifier provided in the onboarding authorization request. The provisioning server is further configured to send an onboarding authorization response to the network function in the onboarding network allowing or rejecting the onboarding authorization request depending on verification of the onboarding identifier.

(15) A ninth aspect of the disclosure comprises a credential server configured to provide network information for a communication network, e.g., SNPN, to an onboarding network to enable provisioning of credentials to the UE to access the communication network. The credential server (e.g., DCS) comprises communicating circuitry for communicating with a network node in an onboarding network over a communication network, and processing circuitry. The processing circuitry is configured to store a binding associating device information for the UE and network information identifying the communication network. The device information includes an onboarding identifier for the UE. The processing circuitry is further configured to receive, from a network function in an onboarding network, a message including the onboarding identifier for the UE. The processing circuitry is further configured to send, to the network function in the

onboarding network, the network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(16) A tenth aspect of the disclosure comprises a network node in an onboarding network configured to obtain network information for a communication network, e.g., SNPN, from a DCS to enable provisioning of credentials to the UE to access the communication network. The network node comprises communicating circuitry for communicating with a credential server over a communication network, and processing circuitry. The processing circuitry is configured to send a message including an onboarding identifier for the UE to a DCS. The processing circuitry is further configured to receive, from the DCS, network information bound to the device information for the UE to enable provisioning of credentials to the UE to access the communication network.

(17) An eleventh aspect of the disclosure comprises a network node in an onboarding network configured to obtain authorization for provisioning of a UE with credentials to access a communication network, e.g., SNPN. The network node comprises communicating circuitry for communicating with a credential server over a communication network, and processing circuitry. The processing circuitry is configured to send, to a network function, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The processing circuitry is further configured to receive an onboarding authorization response from the network function allowing or rejecting the onboarding request. The processing circuitry is further configured to initiate an onboarding procedure when the authorization response indicates that provisioning of the UE is allowed.

(18) A twelfth aspect of the disclosure comprises methods implemented by a provisioning server (e.g., DCS) in a communication network, e.g., SNPN, configured to authorize provisioning of a UE with credentials to access a communication network. The provisioning server comprises communicating circuitry for communicating with a network node in an onboarding network, and processing circuitry. The processing circuitry is configured to receive, from a network function in an onboarding network, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The processing circuitry is further configured to verify the onboarding identifier provided in the onboarding authorization request. The processing circuitry is further configured to send an onboarding authorization response to the network function in the onboarding network allowing or rejecting the onboarding authorization request depending on verification of the onboarding identifier.

(19) A thirteenth aspect of the disclosure comprises a computer program for a credential server. The computer program comprises executable instructions that, when executed by processing circuitry in the credential server causes the credential server to perform the method according to the first aspect.

(20) A fourteenth aspect of the disclosure comprises a computer program for a network node in an onboarding network. The computer program comprises executable instructions that, when executed by processing circuitry in the network node causes the network node to perform the method according to the second aspect.

(21) A fifteenth aspect of the disclosure comprises a computer program for a network node in an onboarding network. The computer program comprises executable instructions that, when executed by processing circuitry in the network node causes the network node to perform the method according to the third aspect.

(22) A sixteenth aspect of the disclosure comprises a computer program for a provisioning server in a communication network. The computer program comprises executable instructions that, when executed by processing circuitry in the provisioning server causes the provisioning server to perform the method according to the fourth aspect.

(23) A seventeenth aspect of the disclosure comprises a carrier containing a computer program

according to the thirteenth aspect. The carrier is one of an electronic signal, optical signal, radio signal, or a non-transitory computer readable storage medium.

(24) An eighteenth aspect of the disclosure comprises a carrier containing a computer program according to the fourteenth aspect. The carrier is one of an electronic signal, optical signal, radio signal, or a non-transitory computer readable storage medium.

(25) A nineteenth aspect of the disclosure comprises a carrier containing a computer program according to the fifteenth aspect. The carrier is one of an electronic signal, optical signal, radio signal, or a non-transitory computer readable storage medium.

(26) A twentieth aspect of the disclosure comprises a carrier containing a computer program according to the sixteenth aspect. The carrier is one of an electronic signal, optical signal, radio signal, or a non-transitory computer readable storage medium.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an exemplary SNPN and an onboarding network configured to facilitate provisioning a UE to access the SNPN.
 - (2) FIG. 2 illustrates an exemplary onboarding procedure for provisioning a UE to access a SNPN.
 - (3) FIG. 3 illustrates a general onboarding procedure for provisioning a UE to access a SNPN.
 - (4) FIG. 4 illustrates an embodiment of the general procedure shown in FIG. 3 providing further details explaining how network information is retrieved from the DCS. In this example, the network information is piggybacked on existing authentication signaling.
 - (5) FIG. 5 illustrates an embodiment of the general procedure shown in FIG. 3 providing further details explaining how network information is retrieved from the DCS. In this example, new signaling is provided to enable the onboarding network to request network information from the DCS.
 - (6) FIG. 6 illustrates an embodiment of the general procedure shown in FIG. 3 providing further details explaining how provisioning is authorized by the SNPN.
 - (7) FIG. 7 illustrates an exemplary method implemented by a DCS for providing network information for a SNPN to an onboarding network to enable provisioning of credentials to the UE to access the SNPN.
 - (8) FIG. 8 illustrates an exemplary method implemented by a network node in an onboarding network for obtaining network information for a SNPN from a DCS to enable provisioning of credentials to the UE to access the SNPN.
 - (9) FIG. 9 illustrates an exemplary method implemented by a network node in an onboarding network to obtain authorization for provisioning of a UE with credentials to access a SNPN.
 - (10) FIG. 10 illustrates an exemplary method implemented by a provisioning server to authorize provisioning of a UE with credentials to access a SNPN.
 - (11) FIG. 11 illustrates an exemplary DCS configured to provide network information for a SNPN to an onboarding network to enable provisioning of credentials to the UE to access the SNPN.
 - (12) FIG. 12 illustrates an exemplary a network node in an onboarding network configured to obtain network information for a SNPN from a DCS to enable provisioning of credentials to the UE to access the SNPN.
 - (13) FIG. 13 illustrates a network node in an onboarding network configured to obtain authorization for provisioning of a UE with credentials to access a SNPN.
 - (14) FIG. 14 illustrates an exemplary method implemented by a provisioning server configured to authorize provisioning of a UE with credentials to access a SNPN.
 - (15) FIG. 15 illustrates a network node configured to facilitate onboarding of a UE in a SNPN.
- ### DETAILED DESCRIPTION

(16) Referring now to the drawings, techniques for provisioning user equipment (UEs) **50** with credentials to access a communication network will be described in the context of a SNPN **100**. In the exemplary embodiment, the SNPN **100** is a 5G network implementing A 5G Core (5GC) **115** and Next Generation Radio Access Network (NG-RAN) **120**. Those skilled in the art will appreciate, however that the techniques are not limited to use in provisioning UEs to access SNPNs **100** or 5G networks but could be applied to enable UEs **50** to access other types of public or private communication networks, both wired and wireless.

(17) FIG. 1 illustrates an exemplary SNPN **100**. The SNPN **100** is a fully functional 5G network comprising a NG-RAN **110** and 5G Core (5GC) **115**. The NG RAN **110** comprises one or more base stations, also known as 5G NodeBs (gNBs) and Next Generation Evolved Node Base station (ng-eNBs), that communicate with UEs **50** authorized by the SNPN **100**. The 5GC **115** comprises a User Plane Function (UPF) **120**, Access and Mobility Management Function (AMF) **130**, Session Management Function (SMF) **140**, Policy Control Function (PCF) **150**, Authentication Server Function (AUSF) **160**, Unified Data Repository (UDR) **170** and Unified Data Management (UDM) Function **180**. The SNPN **100** may further include a provisioning server **190** for provisioning UEs **50** with credentials to access the SNPN **100**. Alternatively, the provisioning server **190** could be a network node external to the SNPN **100** that is controlled by or trusted by the SNPN **100**. The PS **190** in the SNPN **100** communicates with the DCS **75**

(18) Provisioning of UEs **50** with credentials to access the SNPN **100** is facilitated with the aid of an onboarding network **200** and default credential server (DCS) **75**. Generally, the UE **50** initially registers with the onboarding network **200**, which initiates an onboarding procedure with the SNPN **100**. The onboarding network **200** fetches information about the SNPN **100** from the DCS **75** and initiates a provisioning procedure with the SNPN **100**. Once the UE **50** is provisioned with credentials for the SNPN **100**, the UE **500** deregisters from the onboarding network and establishes a connection with the SNPN **100** using the obtained credentials.

(19) The onboarding network **200** may also comprise a 5G network including a NG-RAN **210** and a 5GC **215**. The NG RAN **210** comprises one or more base stations (e.g., gNBs or ng-eNBs) that communicate with UEs **50** authorized by the onboarding network **200**. The 5GC **215** in the onboarding network **200** comprises a UPF **220**, AMF **230**, SMF **240**, PCF **250** and AUSF **260**. The 5GC in the onboarding network **200** may further include a Network Exposure Function (NEF) **270**. The UPF **220** and AMF **230** in the onboarding network **200** communicate with the PS **190** in the SNPN **100** as hereinafter explained in more details. The AMF **230** in the onboarding network **200** also communicates with the DCS **75**.

(20) FIG. 2 illustrates an exemplary onboarding procedure for provisioning a UE **50** with credentials to access the SNPN **100**. The process for onboarding the UE **50** is as follows:

(21) A1. Some minimum configuration in the UE **50** is needed. The UE **50** cannot be provisioned with information of the SNPN **100**, because this would require provisioning or updating its firmware after the UE **50** has been manufactured, but prior to selling it to the SNPN **100**. Rather, the UE **50** is provisioned with a UE identifier, such as a Medium Access Control (MAC) address, host identifier (Host ID), or Permanent Equipment Identifier (PEI), assigned by the DCS **75** and onboarding UE credentials, such as a digital certificate, signed by the DCS **75**, and valid only for onboarding procedures.

(22) A2. Some configuration is also needed in the PS **190**, DCS **75**, and onboarding network **200**. The UE **50** is registered with the DCS **75** by the manufacturer and stores a UE identifier, such as a MAC address, Host ID or PEI. The DCS **75** provides the UE identifier and UE credentials to the onboarding network **200** to enable the UE **50** to access the onboarding network. When the subscription owner (e.g., SNPN owner) acquires one or more UEs **50**, it provisions its PS **190** with the UE identifiers provided by the DCS **75**, along with the data that needs to be provisioned into the UE **50**, such as regular UE credentials for the SNPN **100**, network configuration (network slice information, UE policies, etc.)

(23) B. The UE **50** performs a registration with the onboarding network **200** for the limited purpose of onboarding. The registration is made using a Subscription Concealed Identifier (SUCI) or Subscription Permanent Identifier (SUPI) generated from the UE identifier assigned by the DCS **75**. This SUCI/SUPI is referred to herein as the onboarding identifier and is not useful for any other purpose. The special registration for onboarding includes the authentication of the UE **50** using the onboarding SUCI/SUPI and onboarding UE credentials provided by the DCS **75**.

(24) C. A provisioning procedure is performed to provide the UE **50** with its regular UE credentials and additional configuration for accessing the SNPN **100**. The provisioning procedure may comprise a control plane provisioning procedure or a user plane provisioning procedure.

(25) D. Once the provisioning procedure is completed, the UE de-registers from the onboarding network.

(26) E. The UE **50** uses its regular UE credentials to access the SNPN **100** and establish a Packet Data Unit (PDU) session with the SNPN **100** using regular procedures (registration, PDU session establishment, etc.).

(27) While the procedure shown in FIG. **2** defines the parties involved in the onboarding process, there are many aspects of the onboarding process that remain undefined and/or unresolved. One problem is that the UE **50** is not provisioned in advance with information about the SNPN **100**. Therefore, procedures are needed for network selection and routing of traffic related to provisioning. Further, procedures are needed to prevent rogue or malicious UEs **50** from attempting to access an NPN other than SNPN **100** of the subscription owner. Additionally, the onboarding network **200** needs to determine what procedures to use to provision the UE **50**.

(28) According to one aspect of the present disclosure, the DCS **75** maintains a binding of the UE **50** onboarding credentials and a UE identifier (e.g., MAC address, Host ID, PEI) with network information for the authorized SNPN **100**. Therefore, pre-provisioning of the UE **50** with network information by the manufacturer is not required. The UE **50** sends an onboarding request containing the UE identifier to a network node in an onboarding network **200** and the network node queries the DCS **75** to retrieve the network information. After obtaining the network information from the DCS **75**, the onboarding network **200** requests authorization from a provisioning server **190** in the SNPN **100** to initiate a provisioning procedure with the SNPN **100**. The provisioning server **190** verifies that the UE **50** is authorized to access the SNPN **100** and determines the type of provisioning procedure to use (e.g., control plane provisioning or user plane provisioning). If verification is successful, the provisioning server **190** sends a response authorizing the onboarding network to initiate provisioning of the UE **50** and indicating the type of provisioning procedure to use. The authorization procedure prevents rogue or malicious UEs **50** from attempting to initiate a provisioning procedure with the SNPN without prior authorization.

(29) The binding of the UE credentials and UE identifier with the network information of the SNPN **100** can be defined in the DCS **75** after the UE **50** is manufactured and when the device is sold to a certain SNPN **100**. The network information comprises data identifying the SNPN **100**, such as the PLMN ID and Network Identifier (NID) of the SNPN **100**. It is also possible that the DCS **75** is configured with more than one network owning the SNPN **100** (e.g. each related to a region or different SNPNs have agreements with different onboarding networks **200**) and optionally with information, such as a fully qualified domain name (FQDN), to reach the provisioning server **190** for the SNPN **100**.

(30) After the purchase of a given UE **50** (or most likely a batch of UEs **50**), the SNPN **100** configures its provisioning server **190** with the onboarding SUPIs, which are computed based on the UE identifiers assigned by the DCS **75**. These onboarding SUPIs configured in the SNPN **100** are authorized to subsequently perform onboarding in the SNPN **100**. The configured identities can be complete or part of the onboarding SUPIs, or equivalent information to enable authorization.

(31) The SNPN **100** may also statically allocate a regular SUPI and corresponding credentials for accessing the SNPN **100** to each onboarding SUPI, together with additional data that needs to be

provisioned to the UE **50** at this stage too. Alternatively, the regular UE credentials in the SNPN **100** and additional UE configuration may be dynamically generated by the SNPN **100** at a later stage during the onboarding procedure.

(32) FIG. **3** illustrates an onboarding procedure for provisioning a UE **50** to access a SNPN **100** that improves upon the network selection and security issues. The procedure shown in FIG. **3** follows the same general outline as the onboarding procedure in FIG. **2** with the following modifications:

(33) A1: The UE **50** is provisioned with onboarding credentials as previously described.

(34) A2: The DCS **75** and onboarding network **200** are provisioned as previously described. In addition, the DCS **75** stores a binding of the onboarding UE credentials and the UE identifier (e.g., Host ID, MAC address, PEI, etc.) with the network information for the SNPN **100**. The DCS **75** could be operated by the device manufacturer or another third party. The binding can be defined in the DCS **75** after the UE **50** is manufactured and when the device is sold to a certain SNPN **100**. The network information comprises data identifying the SNPN **100**, such as the PLMN ID and NID of the SNPN **100**. It is also possible that the DCS **75** is configured with more than one network owning the SNPN **100** (e.g., each related to a region or different SNPNs **100** have agreements with different onboarding networks **200**) and optionally with information, such as a fully qualified domain name (FQDN), to reach the provisioning server **190** for the SNPN **100**.

(35) A3: The SNPN **100** configures the provisioning server **190** by adding the onboarding SUCI/SUPI to the list of authorized UEs **50**. The onboarding SUCI/SUPI will be used in the authentication of the UE **50** as hereinafter described. In some embodiments, the SNPN **100** provides the provisioning server **190** with a regular SUPI and corresponding credentials for accessing the SNPN **100** along with additional data that needs to be provisioned to the UE during the initial provisioning. Alternatively, the regular UE credentials in the SNPN **100** and additional UE configuration may be dynamically generated by the SNPN **100** at a later stage (i.e. after UE onboarding authorization in C1 or during actual User Plane or Control Plane UE provisioning procedure in step D1 or D2/D3).

(36) B1: The UE **50** performs a registration with the onboarding network **200** for the limited purpose of onboarding. The special registration for onboarding includes the authentication of the UE **50** using the onboarding SUCI/SUPI and onboarding UE credentials provided by the DCS **75**.

(37) B2: The onboarding network **200** retrieves network information from the DCS **75** that identifies the SNPN **100** owning the subscription for the UE **50** that is attempting the onboarding. This includes the PLMN ID and optionally the NID of the SNPN **100**. The NID is not needed in case PLMN ID is unique (e.g., SNPN **100** is actually a PLMN). The PLMN ID and NID is used by the onboarding network **200** to route to the SNPN **100**.

(38) C1: The onboarding network **200** provides the onboarding SUCI/SUPI to the provisioning server **190** for the SNPN **100**. The provisioning server **190** verifies that the onboarding network **200** is trusted for onboarding purposes, and then verifies internally that the identity of the UE **50** is authorized to perform the onboarding procedure. In some embodiments, the SNPN **100** dynamically generates the regular UE credentials in the SNPN **100** after authorizing the onboarding procedure.

(39) C2: The provisioning server **190**, based on UE capabilities, agreements with the onboarding network **200**, and local policy, decides whether a control plane UE provisioning or user plane UE provisioning procedure should be used. The provisioning server **190** informs the onboarding network **200** about the selected UE provisioning method (PM).

(40) C3: A provisioning procedure is performed to provide the UE **50** with its regular UE credentials and additional configuration for accessing the SNPN. The provisioning procedure may comprise a control plane provisioning procedure or a user plane provisioning procedure as instructed by the SNPN **100**. In some embodiments, the SNPN **100** dynamically generates the regular UE credentials in the SNPN **100** during the control plane UE provisioning or user plane UE

provisioning step.

(41) D. Once the provisioning procedure is completed, the UE de-registers from the onboarding network **200**.

(42) E. The UE **50** uses its regular UE credentials to access the SNPN **100** and establish a Packet Data Unit (PDU) session with the SNPN **100** using regular procedures (registration, PDU session establishment, etc.).

(43) FIG. 4 illustrates an exemplary method for retrieving network information from the DCS **75**. In this embodiment, the SNPN information retrieval (B2 in FIG. 3) is collapsed into the UE registration (B1 in FIG. 3) and the network information is piggybacked on existing Extensible Authentication Protocol (EAP) signaling during the registration procedure.

(44) The SNPN information retrieval proceeds as follows:

(45) B1. The UE **50** sends a registration request to the AMF **230** in the onboarding network **200**.

(46) B2. The AMF **230** selects an AUSF **260**.

(47) B3. The AMF **230** sends an authentication request to the selected AUSF **260** with the onboarding SUPI of the UE **50**.

(48) B4. The AUSF **260** selects a DCS **75**.

(49) B5. The AUSF **260** sends an authentication request to the selected DCS **75**.

(50) B6. The DCS **75** and UE **50** perform EAP authentication procedures.

(51) B7. After the DCS **75** authenticates the onboarding SUPI, it sends an authentication response containing the network information (SNPN info) for the SNPN **100**.

(52) B8. The AUSF **260** sends an authentication response containing the network information (SNPN info) to the AMF **230**.

(53) FIG. 5 illustrates an alternate method for retrieving network information from the DCS **75**. In this embodiment, after successful authentication of the UE, the AMF **230** in the onboarding network **200** requests SNPN information from the DCS **75** using a new service exposed by the DCS **75**. Because the DCS **75** is outside the 3GPP domain, the AMF **230** in the onboarding network **200** may use access this service provided by the DCS **75** via a NEF **270**. The SNPN information retrieval proceeds as follows:

(54) B1. The UE **50** registers with the onboarding network **200** using its onboarding SUPI as previously described.

(55) B2, B3. After registration is completed, the AMF **230** sends an information request to the DCS **75** via NEF **270** to request the network information for the UE **50**. The information request contains the onboarding SUPI of the UE **50** and is sent via a NEF **270**.

(56) B4, B5. The DCS **75** answers, via the NEF **270**, the information request. The response includes the SNPN information, such as the PLMN ID and NID if needed.

(57) In the embodiments shown in FIGS. 4 and 5, the DCS **75** may perform a selection of SNPN **100** to be used for the UE **50** if more than one SNPN **100** is authorized. In some embodiments, the selection of the SNPN **100** can be based on the UE location. Different SNPNs **100** can be provided to serve different regions and the SNPN **100** covering the region where the UE **50** is located can be selected. The onboarding network may need to provide location information to the DCS **75**. In other embodiments, the selection of the SNPN **100** is based on business agreements with SNPNs **100**. These business agreements with SNPNs **100** may include, e.g., certain percentage of UEs per partner, or based on the pre-set agreements with SNPN **100**, such as an SNPN has “bought” the UE **50** and is about to put it into use. In other embodiments, the DCS **75** sends the onboarding network **200** a list with the SNPNs **100** that are provided, along with the regions that the SNPNs **100** are intended to serve. The onboarding network **200** selects an SNPN **100** depending on the location of the UE **50**.

(58) FIG. 6 illustrates a procedure for authorizing the onboarding of the UE **50** by the SNPN **100**. The execution of this procedure provides the onboarding network **200** and the SNPN **100** with a verification that the UE **50** is legitimately owned (or at least authorized) by the SNPN **100** and

prevents rogue or malicious UEs **50** from consuming further resources of the onboarding network **200** and the SNPN **100**. If the UE onboarding authorization is successful, the SNPN **100** decides which UE provisioning method applies to the UE **50** and the onboarding network **200**.

(59) The authorization procedure proceeds as follows:

(60) C1. After obtaining the SNPN information from the DCS **75**, the AMF **230** in the onboarding network **200** selects the SNPN **100** (if the DCS **75** provides more than one) and a provisioning server **190** within the selected SNPN **100**. The selection of the SNPN **100** can be based on information provided by the UE **50** (e.g., interaction not shown in the steps in which the UE **50** is provided by available SNPNs **100** and UE **50** selects one) or by selecting an SNPN **100** in the same region. The AMF **230** in the onboarding network **200** uses the PLMN ID and NID of the selected SNPN **100** to discover and select a provisioning server **190** in the selected SNPN **100** (e.g., by using a DNS lookup with a well-known FQDN, including the SNPN **100** domain). For this purpose, the AMF **230** may reuse existing 5GC procedures (i.e., via a Network Repository Function (NRF)). Alternatively, the DCS **75** may also be configured with the addresses/FQDN of the provisioning server **190** per SNPN **100** and provide the information to the AMF **230** in the onboarding network **200**.

(61) C2. The AMF **230** in the onboarding network **200** sends an onboarding authorization request to the provisioning server **190** to request authorization to the SNPN **100** for the UE to proceed with the onboarding procedure. The AMF **230** in the onboarding network **200** makes use of a service-based interface (SBI) service exposed by the provisioning server **190** at the selected SNPN **100**. The interaction between the AMF **230** in the onboarding network **200** and the provisioning server **190** in the SNPN **100** can be based on the roaming architecture via Security Protection Proxies (SEPPs) or based on exposure interfaces via NEF.

(62) C3. The provisioning server **190** at the SNPN **100** checks if the onboarding SUPI is included in the list of authorized UEs to be onboarded at the SNPN **100** using information pre-configured at the SNPN **100** in step (A4).

(63) C4. If the authorization is successful, the SNPN **100** also decides, based on UE capabilities, agreements with the onboarding network **200**, and local policy, whether a control plane UE provisioning or user plane UE provisioning procedure should be authorized.

(64) C5. At this point, the SNPN **100** may trigger any required provisioning action within the SNPN **100** if not already done, e.g., at pre-configuration of onboarding SUPIs in step A4.

(65) C6. The SNPN **100** delivers the authorization decision to the AMF **230** in the onboarding network **200**. If authorization is successful, the AMF **230** in the onboarding network **200** also receives the information about the provisioning method to use (control plane or user plane) together with any additional information required for the execution of the selected provisioning method.

(66) C7. A provisioning procedure is performed to provide the UE **50** with its regular UE credentials and additional configuration for accessing the SNPN. The provisioning procedure may comprise a control plane provisioning procedure or a user plane provisioning procedure as instructed by the SNPN **100**.

(67) In the onboarding procedures shown in FIGS. 3-6, the UE **50** is not configured with an identity of the SNPN **100**. Instead, the identity of the SNPN **100** is learned throughout a secured onboarding procedure. Thus, the device vendor can manufacture the devices and store them until they are sold. Once sold, the device vendor configures its DCS **75** with data (e.g., a certificate) of the SNPN **100** that has acquired them, together with additional data identifying the PLMN, such as the PLMN ID and NID. The advantage is that the device vendor configures a DCS **75**, which is a simpler operation than the alternative of re-flashing the firmware of already stored devices to provision the SNPN data.

(68) The onboarding network **200** is not provisioned with the onboarding SUPI of the UE **50**, which would require direct and permanent contact between the SNPN **100** and each possible onboarding network **200** in the world for letting the onboarding network **200** the onboarding SUPI

of each UE **50** acquired by the SNPN **100**. Instead, the solution proposed here allows the onboarding network **200** to retrieve the identity (PLMN ID and NID) of the SNPN **100** that has acquired the device from DCS **75**.

(69) The early authorization procedure performed between the onboarding network **200** and the SNPN **100** guarantees that non-authorized UEs **50** are denied access to the network as soon as possible, avoiding resource exhaustion attacks in the onboarding network **200**, and avoiding rogue UEs **50** from being provisioned with data that provides access to the SNPN **100**.

(70) The SNPN **100** determines, based on UE capabilities, agreements with the onboarding network **200**, and local policy, the actual method authorized to the UE **50** for the remote UE provisioning of data.

(71) FIG. **7** illustrates an exemplary method **300** implemented by a DCS **75** for providing network information for a SNPN **100** to an onboarding network to enable provisioning of credentials to the UE **50** to access the SNPN **100**. The DCS **75** stores a binding associating device information for the UE **50** and network information identifying the SNPN **100** (block **310**). The device information includes an onboarding identifier for the UE **50**. The DCS **75** further receives, from a network function in an onboarding network, a message including the onboarding identifier for the UE **50** (block **320**). The DCS **75** sends, to the network function in the onboarding network, the network information bound to the device information for the UE **50** to enable provisioning of credentials to the UE **50** to access the SNPN **100** (block **330**).

(72) FIG. **8** illustrates an exemplary method **350** implemented by an AMF **230** or other network node in an onboarding network for obtaining network information for a SNPN **100** from a DCS **75** to enable provisioning of credentials to the UE **50** to access the SNPN **100**. The AMF **230** sends a message including an onboarding identifier for the UE **50** to a DCS **75** (block **360**). The AMF **230** receives, from the DCS **75**, network information bound to the device information for the UE **50** to enable provisioning of credentials to the UE **50** to access the SNPN **100** (block **370**).

(73) FIG. **9** illustrates an exemplary method **400** implemented by an AMF **230** or other network node in an onboarding network to obtain authorization for provisioning of a UE **50** with credentials to access a SNPN **100**. The AMF **230** sends, to a network function (e.g. provisioning server **190**) associated with the SNPN **100**, an onboarding authorization request to authorize provisioning of credentials to the UE **50** to access the communication network (block **410**). The onboarding authorization request includes an onboarding identifier for the UE **50**. The AMF **230** further receives an onboarding authorization response from the network function in the SNPN **100** allowing or rejecting the onboarding request (block **470**). The AMF **230** initiates an onboarding procedure when the authorization response indicates that provisioning of the UE **50** is allowed (block **480**).

(74) FIG. **10** illustrates an exemplary method **450** implemented by a provisioning server **190** to authorize provisioning of a UE **50** with credentials to access a SNPN **100**. The provisioning server **190** receives, from a network function in an onboarding network, an onboarding authorization request to authorize provisioning of credentials to the UE **50** to access the communication network (block **460**). The onboarding authorization request includes an onboarding identifier for the UE **50**. The provisioning server **190** further verifies the onboarding identifier provided in the onboarding authorization request (block **420**) and sends an onboarding authorization response to the network function in the onboarding network allowing or rejecting the onboarding authorization request depending on verification of the onboarding identifier (block **430**).

(75) FIG. **11** illustrates an exemplary DCS **500** configured to provide network information for a SNPN **100** to an onboarding network to enable provisioning of credentials to the UE **50** to access the SNPN **100**. The DCS **500** includes a storing unit **510**, a receiving unit **520** and a sending unit **530**. The various units **510-530** can be implemented by hardware and/or by software code that is executed by one or more processors or processing circuits. The storing unit **510** is configured to store a binding associating device information for the UE **50** and network information identifying

the SNPN **100**. The device information includes an onboarding identifier for the UE **50**. The receiving unit **520** is configured to receive, from a network function in an onboarding network, a message including the onboarding identifier for the UE **50**. The sending unit **530** is configured to send, to the network function in the onboarding network, the network information bound to the device information for the UE **50** to enable provisioning of credentials to the UE **50** to access the SNPN **100**.

(76) FIG. **12** illustrates an exemplary a network node **550** (e.g., AMF **230**) in an onboarding network **200** configured to obtain network information for a SNPN **100** from a DCS **75** to enable provisioning of credentials to the UE **50** to access the SNPN **100**. The onboarding network node includes a sending unit **560** and a receiving unit **570**. The various units **560-570** can be implemented by hardware and/or by software code that is executed by one or more processors or processing circuits. The sending unit **560** is configured to send a message including an onboarding identifier for the UE **50** to a DCS **75**. The receiving unit **570** is configured to receive, from the DCS **75**, network information bound to the device information for the UE **50** to enable provisioning of credentials to the UE **50** to access the SNPN **100**.

(77) FIG. **13** illustrates a network node **600** (e.g., AMF **230**) in an onboarding network **200** configured to obtain authorization for provisioning of a UE with credentials to access a SNPN **100**. The network node includes **600** comprises a sending unit **610** and a receiving unit **620** and an initiating unit **630**. The various units **610-630** can be implemented by hardware and/or by software code that is executed by one or more processors or processing circuits. The sending unit **610** is configured to send, to a network function (e.g., provisioning server **190**) associated with the SNPN **100**, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The receiving unit **620** is configured to receive an onboarding authorization response from the network function in the SNPN **100** allowing or rejecting the onboarding request. The initiating unit **630** is configured to initiate an onboarding procedure when the authorization response indicates that provisioning of the UE is allowed.

(78) FIG. **14** illustrates an exemplary provisioning server **650** configured to authorize provisioning of a UE **50** with credentials to access a SNPN **100**. The provisioning server **650** comprises a receiving unit **660**, a verifying unit **670** and a sending unit **680**. The various units **660-680** can be implemented by hardware and/or by software code that is executed by one or more processors or processing circuits. The receiving unit **660** is configured to receive, from a network function in an onboarding network, an onboarding authorization request to authorize provisioning of credentials to the UE to access the communication network. The onboarding authorization request includes an onboarding identifier for the UE. The verifying unit **670** is configured to verify the onboarding identifier provided in the onboarding authorization request. The sending unit **680** is configured to send an onboarding authorization response to the network function in the onboarding network allowing or rejecting the onboarding authorization request depending on verification of the onboarding identifier.

(79) FIG. **15** illustrates a network node **700** configured to facilitate onboarding of a UE in a SNPN **100**. The network node **700** comprises communication circuitry **720**, processing circuitry **730**, and memory **740**.

(80) The communication circuitry **720** comprises circuitry for communicating with other networked devices over a communication network. The processing circuitry **730** controls the overall operation of the network node **700** and implements the one or more of the procedures as herein described. The processing circuitry **730** may comprise one or more microprocessors, hardware, firmware, or a combination thereof. The processing circuitry **730** is configured to perform the onboarding procedures as herein described.

(81) Memory **740** comprises both volatile and non-volatile memory for storing computer program code and data needed by the processing circuitry **730** for operation. Memory **740** may comprise any

tangible, non-transitory computer-readable storage medium for storing data including electronic, magnetic, optical, electromagnetic, or semiconductor data storage. Memory **740** stores a computer program **750** comprising executable instructions that configure the processing circuitry **730** to implement the procedures and methods as described herein. A computer program **750** in this regard may comprise one or more code modules corresponding to the means or units described above. In general, computer program instructions and configuration information are stored in a non-volatile memory, such as a ROM, erasable programmable read only memory (EPROM) or flash memory. Temporary data generated during operation may be stored in a volatile memory, such as a random access memory (RAM). In some embodiments, computer program **750** for configuring the processing circuitry **730** as herein described may be stored in a removable memory, such as a portable compact disc, portable digital video disc, or other removable media. The computer program **750** may also be embodied in a carrier such as an electronic signal, optical signal, radio signal, or computer readable storage medium.

(82) Those skilled in the art will also appreciate that embodiments herein further include corresponding computer programs. A computer program comprises instructions which, when executed on at least one processor of an apparatus, cause the apparatus to carry out any of the respective processing described above. A computer program in this regard may comprise one or more code modules corresponding to the means or units described above.

(83) Embodiments further include a carrier containing such a computer program. This carrier may comprise one of an electronic signal, optical signal, radio signal, or computer readable storage medium.

(84) In this regard, embodiments herein also include a computer program product stored on a non-transitory computer readable (storage or recording) medium and comprising instructions that, when executed by a processor of an apparatus, cause the apparatus to perform as described above.

(85) Embodiments further include a computer program product comprising program code portions for performing the steps of any of the embodiments herein when the computer program product is executed by a computing device. This computer program product may be stored on a computer readable recording medium.

(86) In the onboarding procedures as described herein provide a mechanism for discovering the network information for a UE **50** and thus avoid the need to re-flash the firmware of the UE **50** after it is acquired. Further, there is no need to provision the onboarding network **200** with the onboarding SUPI of the UE **50**. Instead, the solution proposed here allows the onboarding network **200** to retrieve the identity (PLMN ID and NID) of the SNPN **100** that has acquired the device from DCS **75**.

(87) The early authorization procedure performed between the onboarding network **200** and the SNPN **100** guarantees that non-authorized UEs **50** are denied access to the network as soon as possible, avoiding resource exhaustion attacks in the onboarding network **200**, and avoiding rogue UEs **50** from being provisioned with data that provides access to the SNPN **100**.

Claims

1. A method implemented by a credential server to support provisioning of a User Equipment (UE) in a standalone non-public network (SNPN), the method comprising: storing a binding associating device information for the UE and network information identifying the SNPN, the device information including an onboarding identifier; receiving, from a network function in an onboarding network, an authentication request message requesting authentication of the UE, the authentication request message including the onboarding identifier of the UE; and sending, to the network function in the onboarding network in response to the authentication request message, an authentication response message comprising the network information bound to the device information to enable provisioning of credentials to the UE to access the SNPN.

2. The method of claim 1, wherein the device information further comprises onboarding credentials for authenticating the UE in the onboarding network.
3. The method of claim 1, wherein the credential server stores binding information associating the device information for the UE and network information for multiple standalone non-public networks (SNPNs).
4. The method of claim 3, wherein sending the network information to the network function comprises sending the network information for the multiple SNPNs to the network function.
5. A credential server comprising: processing circuitry and a memory, the memory storing instructions executable by the processing circuitry whereby the credential server is configured to: store a binding associating device information for a User Equipment (UE) and network information identifying a standalone non-public network (SNPN), the device information including an onboarding identifier; receive, from the network function in an onboarding network, an information request message requesting authentication of the UE, the authentication request message including the onboarding identifier for the UE; and send, to the network function in the onboarding network in response to the authentication request message, an authentication response message comprising the network information bound to the device information to enable provisioning of credentials to the UE to access the SNPN.
6. The credential server of claim 5, wherein the device information further comprises onboarding credentials for authenticating the UE in the onboarding network.
7. The credential server of claim 5, wherein the credential server stores binding information associating the device information for the UE and network information for multiple standalone non-public networks (SNPNs).
8. The credential server of claim 7, wherein to send the network information to the network function the credential server is configured to send the network information for the multiple SNPNs to the network function.
9. A method implemented by a credential server to support provisioning of a User Equipment (UE) in a standalone non-public network (SNPN), the method comprising: storing a binding associating device information for the UE and network information identifying the SNPN, the device information including an onboarding identifier; receiving, from a network function in an onboarding network, an information request message including the onboarding identifier of the UE; and sending, to the network function in the onboarding network in response to the information request message, the network information bound to the device information to enable provisioning of credentials to the UE to access the SNPN.
10. The method of claim 9, wherein the device information further comprises onboarding credentials for authenticating the UE in the onboarding network.
11. The method of claim 9, wherein the credential server stores binding information associating the device information for the UE and network information for multiple standalone non-public networks (SNPNs).
12. The method of claim 11, wherein sending the network information to the network function comprises sending the network information for the multiple communication networks SNPNs to the network function.
13. A credential server comprising: processing circuitry and a memory, the memory storing instructions executable by the processing circuitry whereby the credential server is configured to: store a binding associating device information for a User Equipment (UE) and network information identifying a standalone non-public network (SNPN), the device information including an onboarding identifier; receive, from the network function in an onboarding network, an information request message including the onboarding identifier for the UE; and send, to the network function in the onboarding network in response to the information request message, the network information bound to the device information to enable provisioning of credentials to the UE to access the SNPN.

14. The credential server of claim 13, wherein the device information further comprises onboarding credentials for authenticating the UE in the onboarding network.
 15. The credential server of claim 13, wherein the credential server stores binding information associating the device information for the UE and network information for multiple standalone non-public networks (SNPNs).
 16. The credential server of claim 15, wherein to send the network information to the network function the credential server is configured to send the network information for the multiple SNPNs to the network function.
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